# Design Implications

This study found that parents' needs when using Bairn are influenced by the baby's developmental stage, parents' parenting experience, and real-world caregiving contexts. Parents need not only recording functions, but also features that can be found quickly, health and parenting information that can be understood, and support for translating information into practical caregiving actions. At the same time, real-world parenting activities are immediate and unpredictable, requiring parents to continuously coordinate childcare with recording. Therefore, further design optimisation of Bairn should focus on age adaptation, feature discoverability, recording burden, information comprehension, and practical action support.

# 1. Automatically adjust functions according to the baby's age

Research found that babies at different developmental stages have different caregiving needs, and therefore parents' actual feature use also changes. Participants' babies ranged from 2 to 19 months old, with clear differences in usage needs. Younger babies require more frequent feeding and sleep recording, while as babies grow, the importance of some high-frequency records decreases and parents begin to pay more attention to height, weight, and long-term growth. P7 explicitly wanted the interface to adjust automatically according to the baby's age and hide functions that are not relevant to the current stage.

Design implication: The app should not continuously present a fixed set of functions to all parents. Instead, functions should be dynamically adjusted according to the baby's current age and developmental stage.

Specific design suggestions:

1. Automatically display functions relevant to the baby's current stage;
2. Hide or reduce the priority of functions that are not relevant to the current stage;
3. Gradually adjust the priority of functions such as Feeding, Weaning/Meal, Growth, and Development as the baby grows.
4. Highlight the functions most relevant to the baby's current developmental stage on the homepage.

# 2. Support parent-customisable homepages

Some participants felt that the homepage contained too many functions, resulting in a crowded interface and visual burden, and that they often needed to scroll to find the functions they actually used frequently. P1 wanted to be able to pin frequently used functions while hiding low-frequency functions.

Design implication: The homepage should allow parents to adjust the interface according to their actual parenting needs rather than presenting exactly the same feature layout to every user.

Specific design suggestions:

1. Allow users to pin frequently used functions;
2. Provide a personalised homepage based on users' patterns of use;
3. Keep the homepage simple and reduce visual clutter.
4. Allow parents to rearrange the order of functions on the homepage.

# 3. Organise features according to parents' familiar parenting logic

Research found that parents usually search for features according to their own understanding of parenting tasks rather than systematically exploring the app's existing information architecture. For example, P4 expected Weaning to be located under Feeding; P5 initially could not find Weaning; and the location and name of Firsts also made the feature difficult to discover. Theme 5 further indicates that feature discoverability is closely related to whether the app's information architecture matches parents' existing parenting concepts.

Design implication: Feature categories should correspond as closely as possible to parents' actual parenting tasks and mental models rather than only reflecting the system's internal functional structure.

Specific design suggestions:

1. Group related functions according to real parenting tasks;
2. Group functions that serve similar parenting purposes together.
3. Reduce the need for users to explore multiple levels of navigation to find a feature.
4. Place Meal/complementary feeding records under or near the Feeding section.

# 4. Use simple, intuitive, age-appropriate feature names

Research found that feature names directly affect whether users can understand and discover functions. For example, P4 considered “Firsts” too abstract, while P1 felt that “Weaning” could be associated with pureed food for younger babies and therefore did not seem appropriate for a child who was already eating soft solid meals. P1 suggested using “Meal”, considering it clearer and more suitable for children at different ages.

Design implication: Feature names should use familiar and direct language that accurately communicates the function and remains appropriate across developmental stages.

Specific design suggestions:

1.Avoid overly professional or abstract terminology, for example weaning,first
2. Use parenting terms familiar to parents;
3. Check whether feature names remain appropriate across developmental stages;
4. For example, replace potentially ambiguous “Weaning” with the clearer “Meal”.
5. Ensure that feature names directly communicate what users can do within the feature.

# 5. Reduce the operational cost of high-frequency recording functions

Research showed that Feeding, Sleep, and Nappy were among the most frequently used core functions. P2 could quickly complete feeding records, while P1 also considered Feeding, Sleep, and Nappy highly compatible with everyday parenting activities. At the same time, some participants considered certain recording entry points and operations cumbersome.

Design implication: High-frequency functions should be treated as part of everyday parenting routines, with the number of actions required to enter, input, and save records minimised.

Specific design suggestions:

1. Reduce unnecessary taps when recording Feeding, Sleep, and Nappy activities.
2. Add an automatic weight calculation function to reduce manual calculation during weight recording.

# 6. Support delayed and batch recording

Research found that parents cannot always record information immediately after a caregiving activity. When the baby is crying or during night-time feeding, parents usually prioritise caring for the baby and record information later. Some users record certain information immediately during the day and enter other information together at night.

Design implication: The app should not require parents to complete a record immediately after every caregiving activity. It should accommodate the coexistence of immediate, delayed, and batch recording.

Specific design suggestions:

1. Allow multiple records to be completed together;
2. Save incomplete records;
3. Reduce the risk of information loss caused by delayed recording.
4. Provide a simple “Add previous record” entry for delayed recording.

# 7. Use reminders and automation to reduce memory burden

Research found that parents sometimes forget to record because they are busy with other tasks, or forget to stop the Feeding/Sleep Timer. At the same time, existing reminders helped some users avoid missing feeding or sleep times. Theme 3 therefore indicates that reminders and automation can reduce parents' reliance on active memory and manual operations.

Design implication: The app should proactively help parents complete records that are easy to forget rather than relying entirely on parents to remember when to open the app or start and stop recording.

Specific design suggestions:

1. Provide contextual reminders
2. Prompt users about potentially missed records
3. Provide reminders based on existing Feeding and Sleep records.
4. Provide a simple editing process

# 8. Simplify editing and calculation operations

Research found that editing records after completion can also create operational burden. P3, P5, and P7 all mentioned repeated tapping when modifying Feeding/Sleep Timer records. P5 also noted that recording the baby’s weight required manual calculation and suggested automatic calculation of the weight difference.

Design implication: The recording process should not focus only on data entry; it should also reduce the cost of editing, calculating, and maintaining data.

Specific design suggestions:

1. Use sliding interactions for adjusting time or numerical values, such as editing feeding and sleep durations, instead of requiring repeated taps.
2. Simplify the process of correcting inaccurate records;
3. Automatically calculate data where possible;
4. Add an automatic weight calculation function to reduce manual calculation during weight recording.

# 9. Go beyond storing records and help parents understand them

Research found that the value of digital records is not limited to storing information. Parents use Feeding, Sleep, Meal, and Growth records to understand the baby’s daily condition, routines, and longer-term changes. Theme 1 conceptualises this process as recording → reviewing/comparing → understanding the current situation → supporting caregiving management.

Design implication: The app should help parents understand recorded data rather than simply providing data storage.

Specific design suggestions:

1. Provide a daily summary so parents can quickly review feeding, sleep, and nappy records for the day
2. Support viewing and comparing historical records to help parents understand their baby’s daily patterns and changes
3. Highlight the baby’s current growth range in the Growth Chart and explain the meanings of percentiles
4. Provide relevant explanations based on the recorded stool colour and shape

# 10. Improve the efficiency of finding historical records

Although parents considered historical records useful, P2 noted that finding Feeding records from a specific past day required scrolling through the records. P2 therefore wanted a search function and an easier way to switch between dates.

Design implication: The value of accumulated historical data depends on whether parents can quickly locate the information they need.

Specific design suggestions:

1. Provide quick date switching
2. Add a search function

# 11. Use consistent data presentation across different functions

P1 felt that different functions presented data inconsistently. For example, Nappy and Sleep used daily summaries, while Meal displayed the cumulative number of foods tried, making it harder to directly see what the baby ate on a particular day. P1 preferred a more consistent method of presenting data across sections.

Design implication: Although different recording functions contain different types of information, they should share a consistent data-presentation logic so that parents can quickly understand and compare information.

Specific design suggestions:

1. Add a daily- record summary for Meal, allowing parents to quickly view the day’s Meal entries in the same way they check Feeding, Sleep and Nappy records
2. Separate the display of daily records from cumulative food records. This avoids a situation where users can only see the total number of food items tried, but cannot quickly find out what food the baby ate on a given day

# 12. Translate professional health information into information parents can understand

Growth Chart, percentiles, colours, and medical terminology made it difficult for some parents to judge the baby's actual condition. P2 was unsure about the meanings of different colours and percentiles in the Growth Chart, while P7 also found professional terminology difficult to understand and had difficulty relating the information to the baby's own condition.

Design implication: Health information should not simply display professional data; it should help parents understand what the information means and how it relates to their baby.

Specific design suggestions:

1. Growth Chart: Add brief explanations for percentiles, colour- coding and professional terminology, and highlight the baby's current growth range
2. Nappy records: After users select stool colour and consistency, display corresponding brief explanations and follow- up recommendations

# 13. Use images, videos, and step-by-step instructions to support practical actions

This point is strongly supported by both the Code and Theme. P2 noted that the NHS Guides contained a large amount of content and were mainly text-based, without pictures or videos, and therefore did not continue reading. P3 wanted more images and illustrations, while P8 explicitly suggested short videos and step-by-step illustrations for practical tasks such as breastfeeding, burping, CPR, wrapping, bathing, and choking rescue.

Design implication: For practical caregiving knowledge, text alone may not be sufficient to help parents perform the required actions. Information should therefore be transformed from simply explaining what something is into showing parents how to do it.

Specific design suggestions:

1. Break important practical content in NHS Guides into short steps
2. Add step-by-step illustrations for practical actions such as CPR and the baby recovery position
3. Add short-video access for key practical actions

# 14. Provide parenting knowledge tailored to the baby's age

Research found that parents wanted guidance that matched the baby's current developmental stage rather than large amounts of general information. P2 wanted specific advice for different developmental stages; P3 wanted feeding recommendations to change as the baby grows; and P8 felt that content related to complementary foods, gross motor development, and common childhood illnesses was relatively limited for babies over four months.

Design implication: Parenting knowledge should be filtered and organised according to the baby's current age and developmental stage rather than providing the same large body of content to every user.

Specific design suggestions:

1. Organise NHS guides by age/developmental stage
2. Prioritise content most relevant to the current stage on the homepage
3. Provide stage-specific recommendations for functions such as Feeding
4. Expand NHS Guides beyond the current focus on newborns aged 0 – 3 months by adding guidance for babies aged 6 – 24 months, including complementary feeding and other guidance

# 15. Prioritise high-frequency core functions and reduce low-frequency interface occupation

Research found that parents' actual use was highly concentrated on core recording functions such as Feeding, Sleep, and Nappy, while some other functions, including NHS Guides and Parent Mood, were used less frequently. P1 also noted that as the baby grows, some functions become less relevant but continue to occupy homepage space.

Design implication: Interface space should prioritise functions that parents use frequently and that directly support everyday parenting. Low-frequency functions can remain available but should not compete with core functions for primary interface space.

Specific design suggestions:

1. Place Feeding, Sleep, Nappy, Meal, and other high-frequency functions in easily accessible locations
2. Move low-frequency functions to secondary entry points

3. Adjust feature priority according to actual usage frequency